Diamond Prices

- REVIEW
- ANNOTATIONS 1
- HISTORY

Meets Specifications

Greetings Udacity Student,

Congratulations on completing the Diamond Prices Project! 👍🏽

Great job! Lots of insights, very well presented and great description of your findings!

I wish you all the best for the coming projects and I'm sure you will do great! 😃

I have also uploaded a copy of your project with comments in the Annotation section for your reference.

Take care and keep up the good work!

Here some interesting links for your reference regarding linear regression equation:

- https://www.statisticshowto.datasciencecentral.com/probability-and-statistics/regression-analysis/find-a-linear-regression-equation/
- https://www.dummies.com/software/microsoft-office/excel/how-to-use-the-regression-data-analysis-tool-in-excel/
- https://www.rwu.edu/sites/default/files/downloads/fcas/mns/calculating_and_displaying_regression_statistics_in_excel.pdf
- https://www.analyticsvidhya.com/blog/2015/08/comprehensive-guide-regression/

Predicting Diamond Prices

1. **The student has submitted a PDF file for commenting.**
2. **The student has answered *all* the questions and provided a sufficient explanation of how they arrived at their answer.**

Step 1: Understanding the Model

Answer the following questions:

1. According to the linear model provided, if a diamond is 1 carat heavier than another with the same cut and clarity, how much more would the retail price of the heavier diamond be? Why?
   - The **one additional Carat weight** would result in an **additional $8,413 in price**.
   - The formula created by the regression is:
     [Price = -5,269 + 8,413 x Carat + 158.1 x Cut + 454 x Clarity].
   - Thus, that determined that the **coefficient for a carat** is **$8,413**, so for every **increase** in the **Carat weight** the **diamond price** will **increase** by the amount of the **coefficient**.

Awesome! The coefficient for Carat is 8,413, therefore if cut and clarity are the same in the two diamonds, and the first element is a constant, one carat heavier will increase the price by $8,413. That is, for each additional carat, holding the rest of properties equal, the price will increase by the amount of the Carat coefficient. 👍🏽

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2. If you were interested in a **1.5 carat** diamond with a **Very Good cut** (represented by a **3** in the model) and a **VS2 clarity rating** (represented by a **5** in the model), *how much would the model predict you should pay for it?*
   - As we mentioned above, the formula created by the regression is:
     [Price = -5,269 + 8,413 x Carat + 158.1 x Cut + 454 x Clarity].
   - So, now we will plug in the values for the different variables.
     - **Predicted Price** = -5,269 + 8,413 x **1.5** + 158.1 x **3** + 454 x **5**
     - **Predicted Price = $10,094.80**

Awesome! The formula is indeed correct: Price = -5,269 + 8,413 x Carat + 158.1 x Cut + 454 x Clarity. So, entering the value of the corresponding coefficients in the formula we obtain the price for the specified diamond. Price for the specified the diamond in this Model = $10,094.80 👍🏽

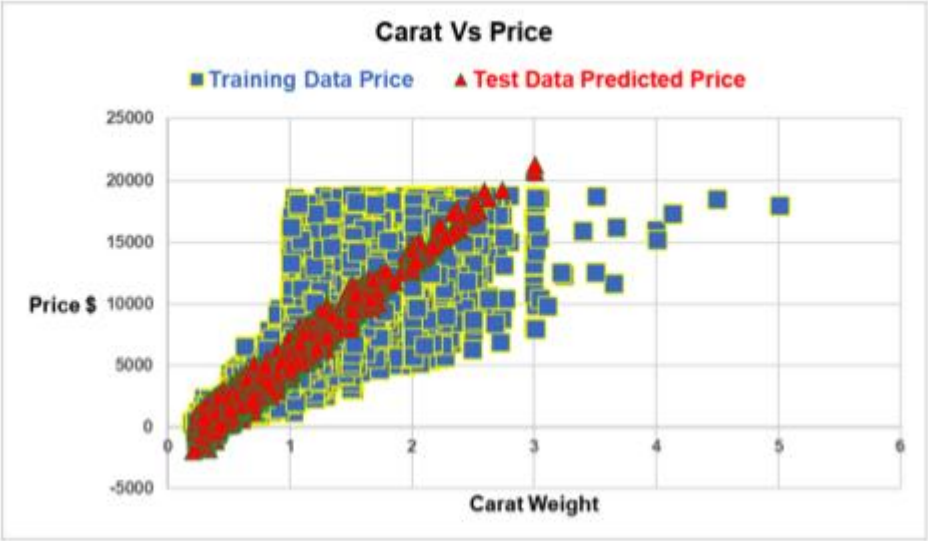
Step 3: Make a Recommendation

Answer the following questions:

1. What price do you recommend the jewelry company to bid? Please explain how you arrived at that number.
   - I **recommend** a bid of **$8,213,465.93**. I arrived at this number by using the given linear equation formula from the regression model provided that was based on **trained** diamond dataset (**50000 records**).
   - I **applied** it to the **attributes** of the **3000 new diamonds** that were up for bid.
   - I then **factored** in the **margin** the Jewelry company were looking for which was **30%** so I multiply the predicted amount **$11,733,522.76** by **.70** to get the final predicted bid of **$8,213,465.93**.

Awesome! Based on the predictive model's findings, and taking into account the jewellery company which generally purchases diamonds from distributors at 70% of the retail price, the formula from the linear regression model (based on the characteristics and prices of the previous diamond sales) get applied to the 3,000 diamonds in order to identify predicted prices. Once this has being computed, the predicted total amount gets multiplied by 0.70 to get the final predicted bid.

1. **The student has provided plots for both old and new diamond prices. The two prices can be plotted in separate plots or a single one.**
2. **The student has defined *carat* on X-axis and *price* on the Y-axis.**



Awesome! You are absolutely right!

Comments about your observations:

- Well done on you observations from the predicted price plot about the trend of the data! Like you mentioned, there may be other factors that are playing an relevant role in the prediction other then the carat variable!
- Well documented analysis! Indeed we have also negative priced in the predicted price dataset hence we need to identify a clear strategy in order to handle these anomalies!
- That's right! Different predicted prices have been identified for same cart weight! We may want to review also the other variables (Cut and Clarity) in order to determine their impact too!
- Agree! Without information such as Adjusted R-Square or P-Value, we have not a able to clear understand model's explained variance nor its accuracy! Also, by looking at the old diamonds dataset, we may even try to build a non-linear model and test its accuracy.

Great Job here! 😃

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